

Candidate Seat No: _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT)

Semester-I University Examination November-2018

CA 403 Information Technology and Digital Electronics

Date: 29.11.18, Thursday Time: 1.30 p.m. to 4.30 p.m. Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.
4. Simple Calculator is permitted.

SECTION-1

Q.1 [A] Fill in the blanks with appropriate answer. [06]

1. Output System of the computer system displays or _____ the results of data processing.
2. _____ is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention.
3. _____ storage type store program and data currently being executed.
4. In the _____ storage cell, read out is destructive.
5. Any kind of commercial transaction that is concluded, over the internet using wireless computing devices is known as _____.
6. _____ type of Intellectual Property covers literary works and artistic works.

[B] Give an appropriate answer to the following multi choice questions. [05]

1. _____ is used to connect to communication network in order to disseminate information in the computer system.
(a) ROM (b) RAM (c) NIU (d) None of these
2. _____ processor is used to process the audio and video data.
(a) Describe Signal Processor (b) Digital Signal Processor
(c) Control Processor (d) None of these
3. The area required to fabricate capacitor storage is _____.
(a) 0.01cm^2 (b) $0.02\mu\text{m}^2$ (c) 0.02cm^2 (d) $0.01\mu\text{m}^2$
4. The data and instructions in _____ are stored by the manufacturer at the time of its manufacturing.
(a) RAM (b) ROM (c) DVD (d) Hard Disk
5. _____ is not the characteristics of IR communication.
(a) Common (b) Expensive (c) Easy to use (d) None of these

Q.2 [A] Differentiate between Data and Information(At least six points). [03]

[B] Discuss the characteristics of storage cell. Explain the Flip flop storage cell in details. [05]

[C] Explain GPS and Bluetooth in details. [04]

OR

Q.2 [A] Give any six societal benefits of using WWW. [03]

- [B] Differentiate between RAM and ROM . Discuss the different types of ROM in details. [05]
- [C] Explain Memory and Processing Systems of Computer Model. [04]

Q.3 Answer following questions in detail (Any Four). [12]

1. Define : Information Technology. Discuss the different types of data with examples.
2. Differentiate SRAM and DRAM.
3. Explain the steps of data processing using computer.
4. Explain Backup, Archival and Secondary storage with characteristics.
5. Define: (a) Virtual Reality (b) Big Data Analytics (c) Internet of Things
6. Discuss the Infrared Communications and Smart Card concepts.

SECTION-2

Q.4[A] Give an appropriate answer from following multi choice questions. [06]

1. What is the Decimal value of an Octal number 226?
(a) 150 (b) 342 (c) 332 (d) 152
2. The Hexadecimal value of Decimal number 567 is _____.
(a) 732 (b) 237 (c) A32 (d) B37
3. NOT gate is also known as _____.
(a) Interrupt (b) Intermediate (c) Inverter (d) None of these
4. The simplification of an equation $(x+y)(x+z)$ is _____.
(a) $x+y+z$ (b) $x + yz$ (c) $y +xz$ (d) $z + xy$
5. If both inputs A and B having 1 value, then what is the output according to Exclusive NOR(EXNOR) gate.
(a) 0 (b) 1 (c) Indeterministic (d) None of these
6. 1 :32 is the variation of _____ digital component.
(a) Multiplexer (b) Decoder (c) Demultiplexer (d) Encoder

[B] **Fill in the blanks with appropriate answer.** [05]

1. Sequential circuit has _____ so output can vary based on input.
2. A _____ is a group of binary cells suitable for holding binary information.
3. The Hexadecimal value of a binary number 10101 is _____.
4. 1's complement of 101110 is _____.
5. The sum terms of $x=1$, $y=0$ and $z=1$ is _____.

Q.5 [A] Explain AND and OR logic gates in details. [04]

[B] Perform the following subtraction using 10's complement : [04]
(1) $345 - 567$ (2) $468 - 234$

[C] Explain 8 :1 Multiplexer with block diagram and truth table. [04]

OR

Q.5[A] Convert following decimal number into a binary number : [04]
(1) 379.25 (2) 865.625

[B] Give the rules of De Morgan theorem and prove them with truth table. [04]

[C] Explain 2 to 4 line decoder with block diagram and truth table. [04]

Q.6

Answer following questions in detail (Any Four).

[12]

1. Perform the following binary multiplications :
(a) 11011100 * 10000001 (b) 1000001 * 11.
 2. Perform the following subtractions using 2's complement :
(a) 11001100 – 11110101 (b) 1111001 - 1011110
 3. Draw the structure of Karnaugh Map based on two,three and four variables.
 4. Explain NAND and EXOR gates with truth table.
 5. Draw the combinational circuit of the following equation :
 $(xyz) + (abc) + (pqr)$
 6. Explain Encoder circuit in details.
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